Axis labels still missing from the thesis figures

Candidates for the same repair as Figures 2.17 and 2.18, with a prompt for each.

Figures 2.17 and 2.18 were survival curves with tick numbers and no axis titles. They have been rebuilt at the width they occupy on the page, with the horizontal title “generation *n*” and the cohort survival probability as the vertical title, at 12 pt for the titles and 10 pt for the tick numbers. This note lists the other quantitative plots in main.pdf with the same defect: an axis, or a colour scale, that shows numbers and does not name the quantity. Photographs, schematics, genealogy trees and cellular-automaton frames were left out. Plots that already name both axes were left out.

Each prompt below is written so it can be pasted to a new agent on its own. The agent should locate the figure in main.pdf, keep the curves, and make the new labels large enough for the width given in the prompt. Where a figure is a raster or a PDF with no editable labels, regenerating it is the reliable route, as it was for Figures 2.17 and 2.18.

The thirteen candidates

High means the figure is hard to read without the label, or the existing label names the wrong axis. Medium means one axis or one panel is unnamed and the caption carries the meaning. Low means the gap is real and small, or the figure is marked as an aside.

#	Figure	Page	Priority	What is missing
1	2.12	45	**High**	Both axes. Tick numbers 0 to 1 only.
2	3.2	66	**High**	Vertical axis says *n*; horizontal axis is untitled.
3	8.6 and 8.7	217	**High**	Upper bar charts. Ticks 0 and 1 only.
4	6.6	163	**High**	Right-hand panels have no vertical title.
5	2.15 and 2.19	48, 52	**Medium**	Vertical axis. Horizontal axis is *t*.
6	3.1	64	**Medium**	Vertical axis empty; ticks are tiny on a narrow panel.
7	3.5	72	**Medium**	Three logarithmic vertical axes.
8	4.2, 4.7, 4.9, 4.13	93–109	**Medium**	One panel in each figure has no vertical title.
9	7.12 (a), (b)	206	**Medium**	Heatmaps with no colour bar.
10	7.5 (a)	186	**Low**	Horizontal axis. Vertical axis is *f(G)*.
11	2.6 (a)	34	**Low**	Horizontal axis of the forcing panel.
12	3.6	73	**Low**	Vertical axis title is empty. Bars are annotated.
13	3.4	69	**Low**	Aside. Both panels have no axes. Not used in the argument.

I did not remeasure every figure that already has axis titles to see whether those titles shrink too far on the page. The size requirement in each prompt is there so a repair does not trade a missing label for an illegible one.

1. Figure 2.12, both axes untitled

High priority. Page 45. File chapters/ch02/figures/dtctA.png, included at 0.48 of the text width.

This is the closest match to Figures 2.17 and 2.18. The plot of the continuous-time constant (green) against the numerical values of the discrete-time constant (red points), with a black regression curve, has tick marks from 0 to 1 on both axes and no axis titles. The caption already says the horizontal quantity is p and the vertical quantity is the constant: green is the continuous-time constant, red is $A(p)$. At half the text width, any label added inside a large default canvas will shrink to unreadability. Draw the replacement at the printed width.

Prompt

Can you locate Figure 2.12 in main.pdf? I want you to add axis labels to it. The horizontal axis is p. The vertical axis is the constant being plotted: the green curve is the continuous-time constant and the red points are the discrete-time constant $A(p)$, which is what the caption already says. This might mean regenerating the figure exactly in Python and adding the labels. The figure is included at 0.48 of the text width, from chapters/ch02/figures/dtctA.png, so the labels need to be large enough to read at that size on the page. Keep the curves, the colours, the grid, and the range of the axes as they are.

2. Figure 3.2, the vertical axis is named with the horizontal quantity

High priority. Page 66. File chapters/ch03/pictures/conditionalMean.pdf, included at 0.74 of the text width.

The curves are the conditional mean $(2p)^n/S_n$ against generation, for p from 0.3 up to 0.49. The vertical axis is labeled n, and the horizontal axis, which runs from 0 to 100, has no title at all. Generation number is the horizontal quantity. The vertical quantity is the conditional mean. A title sentence sits above the frame and does not replace either axis title.

Prompt

Can you locate Figure 3.2 in main.pdf? The vertical axis is labeled n, and the horizontal axis has tick numbers from 0 to 100 and no title. The curves are the conditional mean $(2p)^n/S_n$ against generation number n, for p = 0.3, 0.4, 0.45, 0.48, 0.49, which is what the caption says. Please put generation n on the horizontal axis and the conditional mean on the vertical axis, and remove the n that is currently on the vertical axis. This might mean regenerating the figure exactly and adding the labels. It is included at 0.74 of the text width, from chapters/ch03/pictures/conditionalMean.pdf, so the labels need to be large enough to read at that size. Keep the five curves and their colours.

3. Figures 8.6 and 8.7, the upper bar charts have no axis titles

High priority. Page 217. Files chapters/ch08/figures/repsupp_N20.pdf (full text width) and repsupp_N10.pdf (0.92 of the text width).

Each figure is three columns, for three values of the suppression rate. The lower panel of each column is a heatmap and is already labeled (granule store on the vertical axis, incident replicators on the horizontal axis, survival probability on the colour bar). The upper panel is a bar chart of the full-store marginal. Its vertical axis shows only the ticks 0 and 1, and neither axis has a title. The caption says that upper panel is the survival probability at a full store. Three of those bars sit side by side across the text width, so each chart is narrow and the labels have to be sized for that.

Prompt

Can you locate Figures 8.6 and 8.7 in main.pdf? I want you to add axis labels to the upper bar charts. The lower heatmaps are already labeled; leave them. Each upper chart is the survival probability at a full store, plotted against the initial number of replicators, which is what the caption says. The vertical axis currently shows only the ticks 0 and 1. This might mean regenerating those panels and adding the labels. Figure 8.6 is the full text width and Figure 8.7 is 0.92 of the text width, with three columns in each, so each bar chart is narrow and the labels need to be large enough to read at that size. The files are chapters/ch08/figures/repsupp_N20.pdf and repsupp_N10.pdf. Keep the values, the three suppression rates, and the heatmaps as they are.

4. Figure 6.6, the right-hand panels have no vertical axis title

High priority. Page 163. File chapters/ch06/figures/overlay_V.pdf, included at 0.94 of the text width.

The figure is a two-by-two comparison of free particles (top) and infected cells (bottom), supercritical on the left and subcritical on the right. The left panels are titled, free particles *V(t)* and infected cells *I(t)*, and both columns have the horizontal title time *t*. The right-hand panels have tick numbers only: 0, 2, 4 on the top and 0 to 1 on the bottom. Those are the same two quantities on the subcritical side. The top-left panel is on a logarithmic vertical scale and the top-right panel is linear, so the title should name the quantity and should not imply that the two scales match.

Prompt

> Can you locate Figure 6.6 in main.pdf? The left panels are labeled free particles *V(t)* and infected cells *I(t)*, and both columns have time *t* on the horizontal axis. The two right-hand panels, the subcritical case, have tick numbers and no vertical axis title. Please give them the same vertical titles as the left-hand panels: free particles *V(t)* on the top and infected cells *I(t)* on the bottom. This might mean regenerating the figure. It is included at 0.94 of the text width as a two-by-two layout, from chapters/ch06/figures/overlay_V.pdf, so each panel is about half the text width and the labels need to be large enough to read at that size. Keep the curves, the log scale on the top left, and the linear scale on the top right.

5. Figures 2.15 and 2.19, no vertical axis title

Medium priority. Pages 48 and 52. Files chapters/ch02/figures/abs2.pdf and abs1.pdf, each included at 0.58 of the text width.

Both are state-probability curves against time. The horizontal axis is already titled *t*. The vertical axis has a single tick at 1, an arrow, and no title. The curves are identified in a legend ($p_{0,1}(t)$ and the other states) rather than on the axis. The quantity on the vertical axis is a probability.

Prompt

> Can you locate Figures 2.15 and 2.19 in main.pdf? I want you to add a vertical axis label to each of them. The horizontal axis is already titled *t*. The vertical axis is the state probability: for Figure 2.15 the curves are $p_{0,1}$, $p_{1,0}$ and $p_{0,0}$, and for Figure 2.19 they are $p_{0,1}$ and $p_{1,0}$, which is what the captions say. This might mean regenerating the figures exactly and adding the label. Each is included at 0.58 of the text width, from chapters/ch02/figures/abs2.pdf and abs1.pdf, so the label needs to be large enough to read at that size. Keep the curves, the colours, and the legend.

6. Figure 3.1, empty vertical axis and tiny tick labels

Medium priority. Page 64. Drawn in TikZ from the style myplotstyle in macros/ch03.tex. Three panels, each 0.325 of the text width.

Each panel plots the survival map against the diagonal. The horizontal axis is titled *S*. The vertical axis title is set to an empty string in myplotstyle, and the tick labels are \tiny. On a panel about a third of the text width, those ticks are the same kind of problem as the unlabeled Figures 2.17 and 2.18: the numbers do not survive the printed size. The style is used only by these three panels.

Prompt

> Can you locate Figure 3.1 in main.pdf? Each of the three panels has a horizontal axis titled *S* and no vertical axis title. The tick labels are set to \tiny in the plot style myplotstyle in macros/ch03.tex, and each panel is only 0.325 of the text width, so the numbers are too small on the page. The curves are the map *S* mapped to 2*pS* minus *pS* squared, against the diagonal, for $p = 0.3$, 0.5 and 0.74. Please add a vertical axis title naming that map, and enlarge the tick labels so they are clear at the printed size of the panel. Keep the three curves, the diagonal, and the fixed points.

7. Figure 3.5, three logarithmic vertical axes with no titles

Medium priority. Page 72. File chapters/ch03/pictures/power_law_fixed.pdf, included at 0.98 of the text width.

The horizontal axes are titled (generation n, extinction time, total progeny). Each vertical axis is a logarithmic scale of powers of ten and has no title. Panel titles name the distributions, and the caption gives the three probabilities: survival S_n, the extinction-time mass function, and the total-progeny mass function. Three panels across the text width leaves each axis short, so the new titles have to be sized for a panel about a third of the page.

Prompt

> Can you locate Figure 3.5 in main.pdf? I want you to add vertical axis labels to the three panels. The horizontal axes are already titled. The vertical axes are logarithmic and show only powers of ten. From the caption, panel (a) is the survival probability S_n, panel (b) is the probability mass function of the extinction time, and panel (c) is the probability mass function of the total progeny. This might mean regenerating the figure. It is included at 0.98 of the text width as three panels, from chapters/ch03/pictures/power_law_fixed.pdf, so each panel is about a third of the text width and the labels need to be large enough to read at that size. Keep the curves, the reference lines, and the logarithmic scales.

8. Four Chapter 4 panels with an unnamed vertical axis

Medium priority. Figures 4.2 (page 93), 4.7 (page 100), 4.9 (page 103) and 4.13 (page 109).

These are separate figures, and only one panel of each is missing a title. Figure 4.2 is two stacked path plots: the load X_t is titled, and the released count W_t below it has ticks 0, 10, up to 50 and no vertical title (file F4_paths_W.jpg, 0.72 of the text width). Figure 4.7 panel (b) has ticks from 0 to 1 and no vertical title; panel (a) is titled (file F4_multi_founder.pdf, 0.94 of the text width, two panels). Figure 4.9 has a titled lower panel $I(t)$ against t; the upper panel, which shows $V(t)$ and $J(t)$, has ticks 0 to 12, a legend, and a line of text above the frame, and no vertical axis title (file F4_V_of_t.pdf, 0.72 of the text width). Figure 4.13 panel (a) is titled “Mean value”; panel (b) has ticks up to 17.5 and no vertical title (file F4_conditional_mean.pdf, 0.94 of the text width, two panels).

Prompt

> Can you locate Figures 4.2, 4.7, 4.9 and 4.13 in main.pdf? I want you to add the missing vertical axis labels, and only those. In Figure 4.2 the upper paths are the load and are already labeled; the lower paths are the released count W_t, with ticks and no vertical title. In Figure 4.7, panel (a) is already labeled; panel (b) has ticks from 0 to 1 and no vertical title, and both curves are probabilities, as the caption explains. In Figure 4.9 the lower panel is already labeled $I(t)$ against t; the upper panel shows the expected release $V(t)$ and the expected load $J(t)$ and has no vertical axis title. In Figure 4.13, panel (a) is titled “Mean value”; panel (b) has no vertical title and should be titled in the same way, for the conditional means in that panel. This might mean regenerating each figure. They sit between about 0.72 and 0.94 of the text width, and the two-panel figures are about half of that per panel, so the labels need to be large enough to read at that size. The files are in chapters/ch04/figures/: F4_paths_W.jpg, F4_multi_founder.pdf, F4_V_of_t.pdf and F4_conditional_mean.pdf. Keep the curves and the panels that are already labeled.

9. Figure 7.12, panels (a) and (b) have no colour scale

Medium priority. Page 206. Files chapters/ch07/figures/fig_bdc_three_dists/panel_at_catastrophe.png and panel_no_catastrophe.png, each included at the full text width.

Both panels are heatmaps. The axes are already titled (type-1 count and type-2 count). The colour encodes the probability and has no colour bar and no title. Panel (c), the active distribution, has a colour bar labeled logarithm of the probability. The caption says (a) is the pre-catastrophe distribution at the catastrophe time and (b) is the distribution conditional on no catastrophe. A missing colour bar is the same defect as a missing axis title: the scale that carries the quantity is unnamed. The panels are full width, so there is room for a colour bar with a readable label.

Prompt

> Can you locate Figure 7.12 in main.pdf? Panels (a) and (b) are heatmaps of type-1 count against type-2 count, and the colour has no scale and no label. Panel (c) already has a colour bar labeled with the logarithm of the probability. Please add a colour bar to (a) and to (b), with a label that names the probability the caption assigns to that panel: (a) the distribution of counts at catastrophe, and (b) the distribution conditional on no catastrophe. This might mean regenerating the two panels. Each is included at the full text width, from chapters/ch07/figures/fig_bdc_three_dists/panel_at_catastrophe.png and panel_no_catastrophe.png, so the colour-bar label needs to be large enough to read at that size. Keep the axes, the colours, and panel (c).

10. Figure 7.5, panel (a) has no horizontal axis title

Low priority. Page 186. File chapters/ch07/figures/fig06_type2_phase/fig06.pdf, included at the full text width.

Panel (a) plots *f(G)*, and that vertical title is present. The horizontal axis has ticks from about 0 to 1.4 and no title. The variable is *G*, which panel (c) already uses as a vertical title. Panel (b) is a phase line rather than a cartesian plot. This is one missing title on an otherwise labeled figure, on a panel that shares the page with two others.

Prompt

> Can you locate Figure 7.5 in main.pdf? Panel (a) has a vertical axis titled *f(G)* and a horizontal axis with tick numbers and no title. The horizontal quantity is *G*, the same variable panel (c) already names. Please add that horizontal label. This might mean regenerating the figure. It is included at the full text width as three panels, from chapters/ch07/figures/fig06_type2_phase/fig06.pdf, so the label needs to be large enough to read on that panel. Leave panels (b) and (c) as they are, and keep the curves.

11. Figure 2.6, panel (a) has no horizontal axis title

Low priority. Page 34. File chapters/ch02/figures/seasonal_lv.pdf, included at 0.78 of the text width.

Panels (b) and (c) share a horizontal axis, and “time” is printed under panel (c), which is the right place for a stacked pair. Panel (a) is a different horizontal scale, ticks from 0 to 2, one or two forcing periods, with the vertical axis titled Season(*t*) and no horizontal title. The caption describes it as the clipped seasonal forcing.

Prompt

> Can you locate Figure 2.6 in main.pdf? Panels (b) and (c) already share a horizontal title, time. Panel (a), the seasonal forcing, has a vertical axis titled Season(*t*) and a horizontal axis with ticks from 0 to 2 and no title. Please add a horizontal axis label for that panel, naming the time in the units the panel already uses. This might mean regenerating the figure. It is included at 0.78 of the text width, from chapters/ch02/figures/seasonal_lv.pdf, so the label needs to be large enough to read at that size. Keep the three panels and their curves.

12. Figure 3.6, the vertical axis title is an empty string

Low priority. Page 73. Drawn in TikZ in chapters/ch03/chapter.tex, width 14 cm. The figure has no label command in the source; it is numbered 3.6 by position.

The horizontal axis is titled *n*. The vertical axis title is set to an empty group. The blue bars are the terms 1, 1/2, 1/3, and so on, and each bar carries its own fraction, so a reader can recover the quantity, but the axis itself is unnamed. The caption is about the divergence of the harmonic series. This is a smaller gap than the others because the terms are written on the bars.

Prompt

> Can you locate Figure 3.6 in main.pdf, the harmonic-series figure on page 73? The horizontal axis is titled *n* and the vertical axis title is empty in the TikZ in chapters/ch03/chapter.tex. The bar heights are 1/*n*. Please add a vertical axis label naming that term. The figure is about 14 cm wide, so the label should be large enough to read at that size, in line with the surrounding text. Keep the bars, the curve, and the annotations.

13. Figure 3.4, an aside whose two panels have no axes

Low priority. Page 69. File chapters/ch03/pictures/period_double.png, included at 0.8 of the text width. The caption begins "Aside, not used in the argument."

The left panel is a bifurcation diagram with no ticks and no axis titles. The right panel is a cobweb on a grid, again with no ticks and no axis titles. The caption says the left panel is sampled at $r = 3.5255$ and the right panel is a trajectory of the logistic map at that value, and that the branching window sits at the far left of the diagram. I would leave this unless you want the aside held to the same standard as the other plots. It is not used in the argument.

Prompt

> Can you locate Figure 3.4 in main.pdf? The caption marks it as an aside, not used in the argument. Both panels have no axis titles and no tick numbers. The left panel is the logistic-map bifurcation diagram, with *r* horizontal and the attractor vertical, and the right panel is a cobweb of the map at $r = 3.5255$. Please add those axis labels, large enough to read: the figure is included at 0.8 of the text width, from chapters/ch03/pictures/period_double.png. This might mean regenerating the figure. Keep the diagram and the cobweb as they are, and do not pull this figure into the argument.

Seen, and not given a prompt

Three matplotlib sketches in chapters/ch10/sections/02_two_phase.tex (rough.png, rough2.png, rough3.png) have the same bare axes as Figure 2.12: tick numbers only, on a small raster. That section is not input by the compiled Chapter 10, so the sketches are not in main.pdf. Three birth–death trajectory images in chapters/ch01/figures/fig_bd_regimes/ are likewise untitled and are not included. I left both sets out of the prompts.

Stacked figures that name the shared axis on the bottom panel only, such as the time axis under Figure 2.6 panels (b) and (c), were treated as labeled. Panel (a) of that figure is candidate 11 because its horizontal scale is not the one titled underneath.